Sheet 1 Microcontroller

Multiple Choice Questions

Choose the correct option for each question.

1. The 8051 was developed by:

a) Motorola
b) Intel
c) Zilog
d) Microchip

2. The 8051 follows which architecture?

a) Von Neumann
b) Harvard
c) RISC only
d) Pipeline

3. The on-chip program ROM of the standard 8051 is:

a) 1K b) 2K
c) 4K d) 8K

4. The internal RAM of the 8051 is:

a) 64 bytes
b) 128 bytes
c) 256 bytes
d) 4K bytes

5. How many 8-bit I/O ports does the 8051 have?

a) 2
b) 3
c) 4
d) 8

6. The 8051 has how many 16-bit timers/counters?

a) 1
b) 2
c) 3
d) 4

7. The 8051 is packaged as a DIP with how many pins?

a) 28
b) 32
c) 40
d) 44

8. Which pin supplies the +5V power?

a) Pin 20
b) Pin 31
c) Pin 9
d) Pin 40

9. The GND pin of the 8051 is pin number:

a) 18
b) 19
c) 20
d) 21

10. Port 0 occupies pins:

a) 1–8
b) 10–17
c) 21–28
d) 32–39

11. Which port acts as the multiplexed low-byte address/data bus?

a) P0
b) P1
c) P2
d) P3

12. Which port provides the high byte of the address for external memory?

a) P0
b) P1
c) P2
d) P3

13. Which port is used only as general-purpose I/O?

a) P0
b) P1
c) P2
d) P3

14. P3.0 and P3.1 are the alternate-function pins for:

a) INT0, INT1
b) T0, T1
c) RD, WR
d) RXD, TXD

15. The external interrupt 0 (INT0) is on:

a) P3.1
b) P3.2
c) P3.3
d) P3.4

16. P3.6 and P3.7 serve as:

a) Timer inputs
b) Serial pins
c) WR and RD strobes
d) Crystal pins

17. ALE stands for:

a) Address Load Enable
b) Address Latch Enable
c) Array Latch Enable
d) Access Latch Enable

18. The ALE signal is used to latch the address output at:

a) Port 1 and Port 3
b) Port 0 and Port 2
c) Port 1 only
d) Port 3 only

19. PSEN is the read signal for:

a) External data memory
b) Internal RAM
c) External program memory
d) I/O ports

20. For an 8031 (no on-chip ROM), the EA pin must be connected to:

a) Vcc
b) GND
c) RST
d) XTAL1

21. For an 8051 using its internal ROM, the EA pin is connected to:

a) GND
b) Vcc
c) ALE
d) PSEN

22. The RST pin is:

a) Active low
b) Active high
c) Output only
d) Edge-triggered only

23. The RST pulse must stay high for at least:

a) 1 machine cycle
b) 2 machine cycles
c) 4 machine cycles
d) 8 machine cycles

24. The typical capacitors used with a quartz crystal at XTAL1/XTAL2 are:

a) 3 pF
b) 30 pF
c) 300 pF
d) 3 µF

25. When an external TTL oscillator is used, XTAL2 is:

a) Grounded
b) Connected to Vcc
c) Left unconnected
d) Connected to XTAL1

26. How many register banks does the 8051 have, and what registers does each contain?

a) 2 banks, R0–R3
b) 4 banks, R0–R7
c) 4 banks, R0–R15
d) 8 banks, R0–R7

27. Register Bank 2 occupies RAM addresses:

a) 00H–07H
b) 08H–0FH
c) 10H–17H
d) 18H–1FH

28. RS1 = 1 and RS0 = 1 in the PSW selects:

a) Bank 0
b) Bank 1
c) Bank 2
d) Bank 3

29. The bit-addressable RAM area spans:

a) 00H–1FH
b) 20H–2FH
c) 30H–7FH
d) 80H–FFH

30. Special Function Registers occupy addresses 80H–FFH and are accessed using:

a) Indirect addressing only
b) Direct addressing
c) Register banks only
d) Immediate addressing only

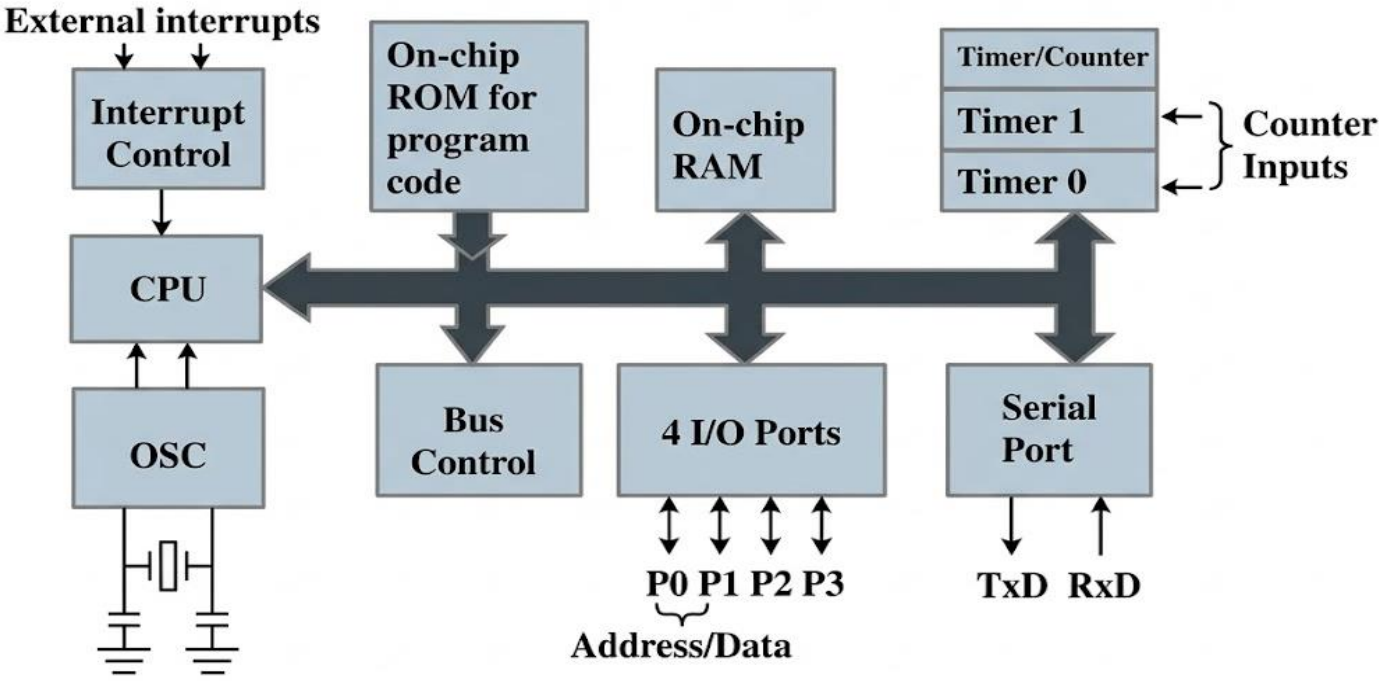
Answer Key (MCQs)

1	**b**	**11**	**a**	**21**	**b**
2	**b**	**12**	**c**	**22**	**b**
3	**c**	**13**	**b**	**23**	**b**
4	**b**	**14**	**d**	**24**	**b**
5	**c**	**15**	**b**	**25**	**c**
6	**b**	**16**	**c**	**26**	**b**
7	**c**	**17**	**b**	**27**	**c**
8	**d**	**18**	**b**	**28**	**d**
9	**c**	**19**	**c**	**29**	**b**
10	**d**	**20**	**b**	**30**	**b**

31. Differentiate between a microprocessor and a microcontroller with suitable examples.

Comparison Parameter	Microprocessor	Microcontroller
Integration Level	CPU is stand-alone; RAM, ROM, I/O, timers are separate	CPU, RAM, ROM, I/O ports, and timers are integrated on a single chip
Component Flexibility	The designer can decide on the amount of ROM, RAM, and I/O ports	Fixed amount of on-chip ROM, RAM, and I/O ports
Clock Speed / Frequency	High speed (typically 1 GHz – 4 GHz+)	Moderate to low speed (typically 8 MHz – 300 MHz)
Power Consumption	High power consumption (requires external cooling & power rails)	Low power consumption (optimized for low-power idle/sleep modes)
Cost	Higher overall system cost (due to external peripherals & complex PCB layout)	Lower total system cost (compact single-chip solution)
Application Focus	General-purpose computing (PCs, laptops, servers)	Single-purpose / Dedicated embedded control (appliances, automotive, IoT)
System Versatility	High versatility; suited for complex multitasking	Optimized for dedicated, real-time control tasks
Design Priority	High computational throughput & memory scalability	Critical for applications where cost, power, and space are tight

32. Draw and explain the block diagram of the 8051 microcontroller.



33. List the basic components and features of the 8051 (ROM, RAM, I/O ports, timers, serial interface).

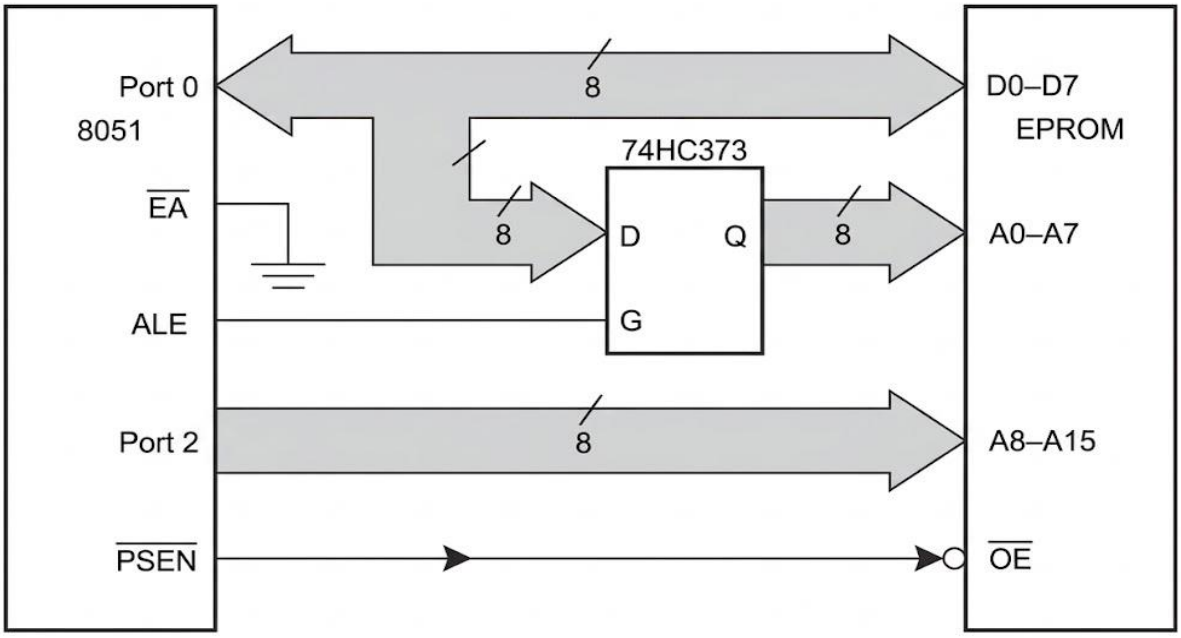
1) 4K bytes internal ROM
2) 128 bytes internal RAM
3) Four 8-bit I/O ports (P0 to P3)
4) Two 16-bit timers/counters
5) One serial interface (UART)
6) 40-pin package, +5V supply

34. Explain the functions of the four I/O ports (P0 to P3) of the 8051, with their PINs

Each port is 8 bits wide and can be used as input or output (bidirectional).

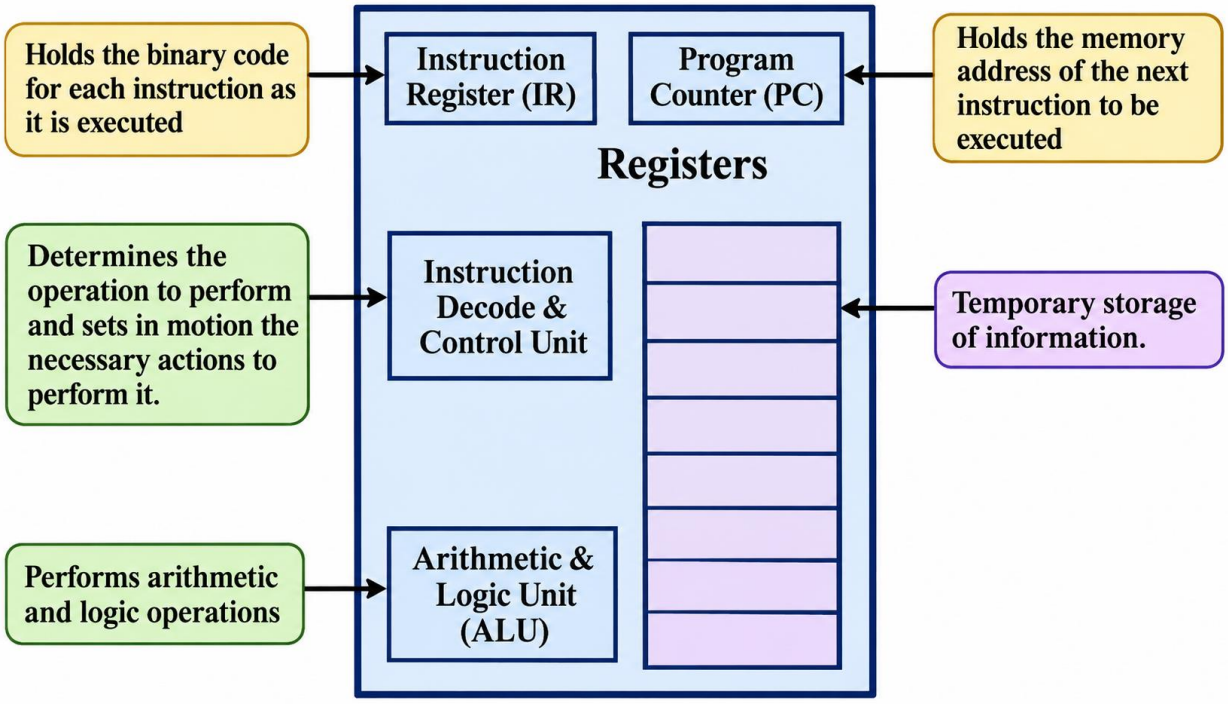
1) **Port 0 (pins 32 to 39, P0.0 to P0.7):** general-purpose I/O, or the multiplexed low-byte address/data bus (AD0 to AD7) for external memory.
2) **Port 1 (pins 1 to 8, P1.0 to P1.7):** general-purpose I/O only.
3) **Port 2 (pins 21 to 28, P2.0 to P2.7):** general-purpose I/O, or the high byte of the address bus (A8 to A15) for external memory.
4) **Port 3 (pins 10 to 17, P3.0 to P3.7):** general-purpose I/O if the internal peripherals (timers) and external interrupts are not used.

35. Draw the Address Multiplexing for External Memory



.

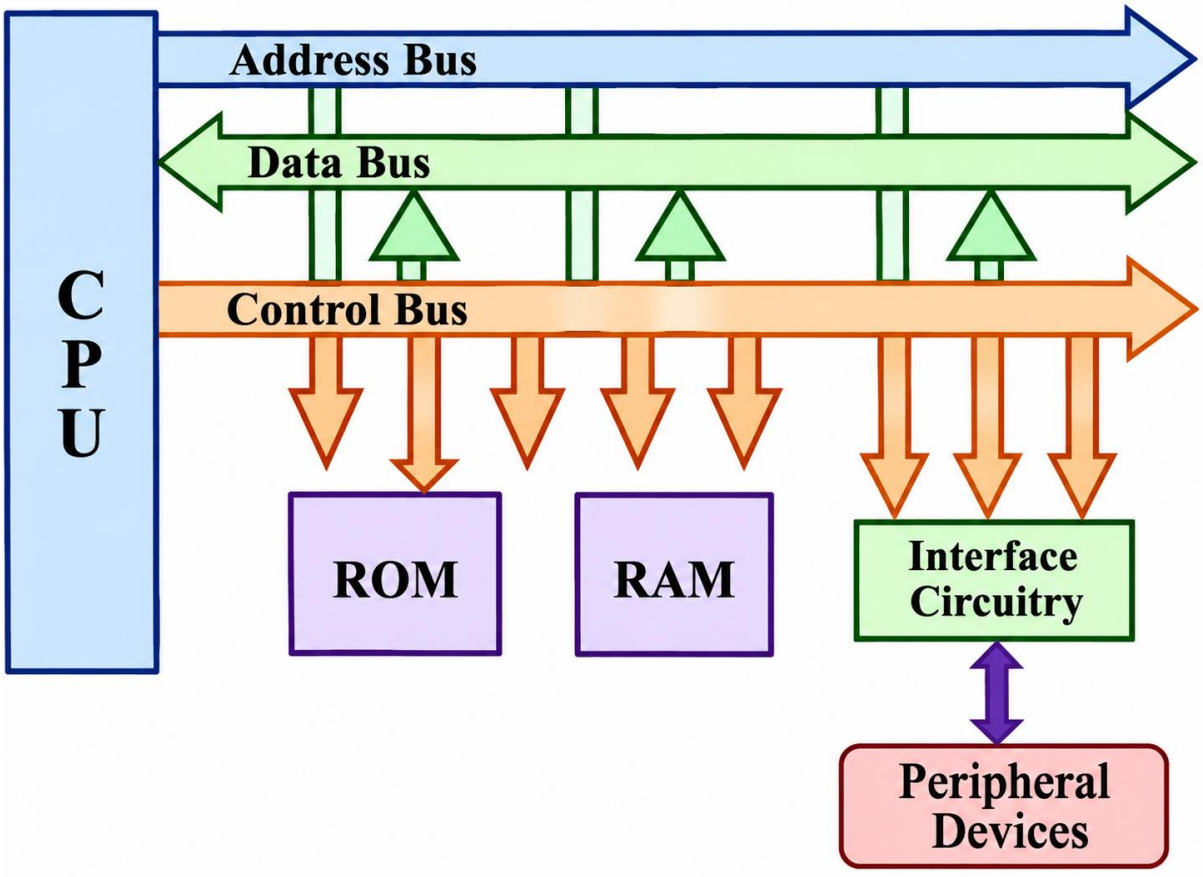
36. Draw the Block Diagram of Microprocessor



37. Write the alternate functions of the Port 3 pins.

Pin	Alternate function
P3.0	RXD (serial input)
P3.1	TXD (serial output)
P3.2	INT0 (external interrupt 0)
P3.3	INT1 (external interrupt 1)
P3.4	T0 (Timer 0 external input)
P3.5	T1 (Timer 1 external input)
P3.6	WR (external data memory write strobe)
P3.7	RD (external data memory read strobe)

38. Draw the Block Diagram of Microcomputer



39. Draw the Block Diagram of Microcontroller

